

Evidential Bayesian Neural Networks for Uncertainty-Aware Classification

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TODO: defence date

Showcase
Figure

Outline

- 1 Introduction
- 2 Background
- 3 Methods
- 4 Experiments
- 5 Discussion
- 6 Limitations and Future Work
- 7 Conclusion

A Class Label Alone Is Not Enough

Message. Annotators disagree on some images. A model must say how sure it is, and why.

TODO: content. One CIFAR-10H image with its count vector over the ten classes. One sentence on predictive uncertainty and its sources.

Three Sources of Predictive Uncertainty

Message. Predictive uncertainty has three sources. Aleatoric, epistemic, and distributional uncertainty.

TODO: content. Three rows. Name, one line definition, one intuition each. No equations.

Message. Categorical BNNs represent two of the three terms. EDL represents two. We combine them into two models that represent all three.

TODO: content. Table of models against uncertainty terms. Two lines. Cat-eBNN with one label per input and a prior. DM-eBNN with repeated labels. Spell out all acronyms.

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Message. A BNN has a posterior over the network weights. Prediction averages over it.

TODO: content. Weight posterior $p(w \mid D)$. BMA predictive. Categorical likelihood with $f(x; w)$.

Message. A Dirichlet has a mean vector and a total concentration parameter. Rescaling the concentration parameters changes the total concentration parameter and leaves the mean vector unchanged.

TODO: content. Density, mean vector, total concentration parameter, scale invariance. Small simplex figure with two total concentration parameters at one mean vector.

Message. Predictive uncertainty splits into aleatoric, distributional, and epistemic terms. At a fixed mean vector, the total concentration parameter sets the split between aleatoric and distributional uncertainty.

TODO: content. Three term decomposition in one line. Cat-BNN point mass, zero distributional term. Small figure of both terms against α_0 .

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Message. The Cat-eBNN is EDL's Dirichlet network with a posterior over the shared network weights. It represents all three terms.

TODO: content. Model sketch as a small graph. Likelihood is the Dirichlet mean $m_y(x; w)$.

Message. The single label likelihood is invariant to the total concentration parameter at a fixed mean vector. So we state the assumption with a Gamma prior on $\alpha_0(x; w)$.

TODO: content. Invariance in one line. Gamma prior with prior mode and prior standard deviation. Honest scope sentence.

Message. Repeated labels, aggregated by class into count vectors, are a different observation. The DM-eBNN models them with a Dirichlet multinomial likelihood. The M-BNN is the Bayesian baseline with a multinomial likelihood.

TODO: content. Count vector n and count total R . The two observation models side by side. Spell out M-BNN.

Message. At a fixed mean vector and a count total above one, the count variance depends on the total concentration parameter. So the count likelihood can inform it. The M-BNN has a fixed count variance.

TODO: content. Dirichlet multinomial variance in one line. Variance ratio in one line. The Gamma prior stays part of the posterior.

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Message. We compare the Cat-eBNN with the Cat-BNN on Fashion-MNIST and CIFAR-10, and the DM-eBNN with the M-BNN on CIFAR-10H.

TODO: content. Two columns, single labels and repeated labels. Datasets, models, SGLD with one chain and a fixed budget, initialisation, prior sweeps, metrics, mean matched controls.

Message. The realised total concentration follows the prior mode. A narrower prior keeps it closer. A higher realised total concentration gives a higher aleatoric share and a lower distributional share.

TODO: content. Extracts of the prior mode sweep and the prior sd sweep. One line on the fixed mean effect.

Message. The Cat-eBNN has competitive accuracy but no consistent predictive improvement over the Cat-BNN. Its contribution is the complete uncertainty representation and the explicit assumption.

TODO: content. Extract of the predictive table. One line on the decomposition table.

Message. The DM-eBNN has lower count NLL than the M-BNN under both initialisations and both prior modes, with competitive majority label accuracy. The mean matched controls show that the additional count variance drives the better fit.

TODO: content. Extract of the count prediction table. Rows M-BNN, DM-eBNN, mean matched control.

Message. The count data inform the realised total concentration, but the prior mode and the initial checkpoint still affect it within the sampling budget. Without the Gamma prior, every cold initialisation fit reaches the boundary $\alpha_0 = C$.

TODO: content. Extract of the prior sensitivity table. One panel of the cold collapse figure.

Message. The DM-eBNN reports all three terms, and count data inform the split between aleatoric and distributional uncertainty.

TODO: content. Extract of the count terms table.

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Message. The eBNNs are the only models here that represent all three terms. They also differ in how the split is set.

TODO: content. The table from slide 1.3 with one new column. How the split is set. Point mass, training objective, stated prior, prior and count data.

Message. Making the assumption explicit does not remove the single label limit. Repeated labels do. This motivates collecting and retaining repeated labels.

TODO: content. Three lines. The Cat-eBNN states the assumption. Single labels cannot inform it. Count data can, and also improve count prediction.

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Message. The evidence has three limits.

TODO: content. One SGLD chain per run. One dataset with repeated labels, no simulated data. The practical value of the decompositions is open.

Message. Four directions follow from the limits.

TODO: content. Several chains and other samplers. Simulated data with known label variation. Further datasets. Decision tasks.

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Message. Two evidential BNNs represent all three uncertainty terms. Single labels cannot inform the split, so the Cat-eBNN states it as a prior. Repeated labels can inform it, and the DM-eBNN also predicts count data better than the M-BNN.

TODO: content. Three lines, the thread of STORY.md Section 0.

TODO: content. Bayesian Dirichlet combination models. Models for repeated labels.
Posterior samplers.

TODO: content. Cold, warm, and shifted initialisation. Trajectory figure or table extract.

TODO: content. Aleatoric term in closed form. Distributional term as the remainder.

TODO: content. Scale invariance. Product of mean components. Posterior odds unchanged.

TODO: content. Density, prior mode, prior standard deviation.

TODO: content. Marginal likelihood loss is invariant to $\alpha_0(x; w)$. The KL term sets it.

Backup. SGLD and the Sampling Budget

TODO: content. Sampler, one chain per run, budget, checkpoint restore.

TODO: content. One frame per full table as needed.

TODO: content. Per seed tables for single labels and repeated labels.

Backup. Cold Initialisation without the Gamma Prior

TODO: content. The four panels of the cold collapse figure.